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1 Motion Is Not Meaning: Reduced-Motion Equivalence in Field-Based Interface Systems

> **Status: research draft (preprint, work in progress).** Paper 4 of the Fundamental family — the accessibility validator. Claims verified against the codebase and canonical docs as of 2026-06-07. See the series index and *the caveat canon* therein. This is a preprint draft, not canonical product documentation.

Author: Zach Shallbetter **Series:** Fundamental Research Papers, Paper 4 of 8 (accessibility / reduced-motion equivalence) **Companion papers:** the flagship paradigm paper, Fundamental: A Field Translation Runtime for Relational DOM Interfaces, establishes the model, the six-registry platform, and the truth-mode vocabulary this paper assumes; see also Reading Field (the empirical reading study), Evidence Fields (trust and provenance). See the series index.

1.1 Abstract

Interfaces that express meaning through motion present a dilemma. If motion is expressive enough to be useful, it risks being inaccessible to readers who cannot or should not be exposed to it — readers with

vestibular disorders, readers who set `prefers-reduced-motion`, readers on assistive technology. If motion is dialed back to be safe, it risks being dismissed as decoration that carried no meaning in the first place. The usual escape — a separate "reduced" code path — drifts from the real interface over time and is rarely tested against it.

This paper develops Fundamental's resolution of that dilemma into a checkable property. The single claim is: **expressive field behavior can remain accessible when motion is treated as one *representation* of field state, not the *source* of meaning — and that equivalence can be made a testable conformance property rather than a hand-waved fallback.** We give (a) a formal equivalence model that traces meaning from a *semantic source* (live DOM text and state) through a *visual–semantic binding* to a *motion behavior* and its *static equivalent*; (b) a conformance model that makes "motion is not meaning" mechanically checkable through shipped lint rules, a pattern schema that makes a reduced-motion fallback *required*, and a proposed automated equivalence harness; and (c) an evaluation plan that tests whether the reduced surface is non-inferior to the full-motion surface on comprehension, orientation, and recall. We are precise about what ships — the reduced-motion guard, the travel-gating user agent, the accessibility lint, the recipe-required accessibility field, and the accessibility preview — and what is proposed. We claim no empirical result; the study is a protocol.

1.2 1. Introduction

1.2.1 1.1 The double bind of expressive motion

Motion in an interface is rarely neutral. A transition that travels can show *where a thing came from*; a pulse can show *what is active*; a flowing ribbon can show *which way a relationship points*. Fundamental leans into this: its reciprocal loop (Paper 1, §3) writes field state — attention, heat, memory, relationship strength — back onto the DOM, and a natural way to render those continuous scalars is through movement.

But movement is exactly the channel that the accessibility literature warns about. A non-trivial population experiences motion as a barrier rather than an aid: vestibular disorders can make large-scale or parallax motion nauseating or disorienting, and the platform exposes a first-class signal — the `prefers-reduced-motion` media query — for readers who have asked the system to stop moving. [prefers-reduced-motion; wcag-motion; vestibular-accessibility] An interface whose *meaning* is encoded in motion fails these readers twice: it withholds the meaning *and* it cannot simply turn the motion off without turning the meaning off with it.

The mirror-image failure is just as real. An interface that uses motion *only* decoratively — motion that carries no state, that could be removed with no loss — invites the conclusion that the whole expressive layer is decoration. For a system like Fundamental, which asks designers to reason about *cause* ("why is this emphasized? what is pulling attention?"), being read as a particle toy is an existential risk, not a cosmetic one. The flagship paper names this directly as a threat to the framing: "the risk of 'spectacle over meaning' is real" (Paper 1, §10).

1.2.2 1.2 The antipattern: the degraded second path

The conventional answer to "make it accessible" is a *fallback*: a second, simpler rendering that the system shows when motion is unwelcome. The trouble with a fallback-as-second-path is structural. It is authored separately, so it can encode *different* information from the primary surface; it is exercised rarely, so it drifts; and because it is framed as a lesser experience, the meaning it must preserve is treated as negotiable. A fallback that quietly drops a state the primary surface conveyed is not a smaller version of the interface — it is a different, lossy one. The accessibility canon in this project names the failure mode explicitly: `MISSING_REDUCED_MOTION`, "no accessible fallback" (`packages/core/src/contracts/guards.ts`).

1.2.3 1.3 The Fundamental stance: the same field, differently revealed

Fundamental takes a different stance, stated in the flagship as the overview this paper expands (Paper 1, §8.5): **motion is one representation of state, not the meaning itself.** Under that stance, reduced motion

is not a degraded fallback path but *the same field differently revealed.* The underlying field — attention, memory, relationship, heat — is computed identically whether or not motion is allowed; what changes is only *how that computed state is rendered.* A particle's travel and a frozen highlight are two renderings of one density value. Dropping the travel does not drop the density; it changes the pen, not the number.

This reframe rests on a property the platform already enforces. Field state lives in CSS custom properties and typed registries, not in pixel motion: the canonical accessibility preview puts it plainly — under `prefers-reduced-motion: reduce` "the engine freezes the simulation (`dt = 0`) — but the field's *state* still reads, because it lives in CSS variables, not in travel" (`apps/site/src/pages/docs/accessibility-preview.astro`). The state is the thing; motion is one of its skins.

1.2.4 1.4 Contributions

This paper contributes the accessibility model behind that stance:

1. **A formal equivalence model** (§3): a four-part chain — *semantic source → visual–semantic binding → motion behavior → static equivalent* — with a mapping table from each motion form to its static equivalent, and three invariants the chain must preserve (*meaning is never motion-only*; *color is never the sole carrier*; *state still tracks under reduced motion, only easing drops*).
2. **A conformance model** (§4) — the central contribution — that turns "motion is not meaning" from an aspiration into a checkable property, through shipped lint rules (`visual-orphan`, `visual-not-hidden`, `feedback-non-css-var`), a pattern schema that makes a reduced-motion fallback *required* (`accessibility.reducedMotion / meaningWithoutMotion`), a reduced-motion guard, a travel-gating user agent, and a *proposed* automated equivalence-conformance check, with a precise Implementation-status accounting of ships-versus-proposed.
3. **An evaluation plan** (§5): a non-inferiority study design that asks whether the reduced surface preserves meaning — comprehension, orientation, recall — across a full-motion, reduced-motion, and static-baseline condition, with a preference probe for motion-sensitive readers and a probe for any meaning a reader can get *only* from motion.

Scope discipline. This paper concerns accessibility and reduced-motion equivalence *only.* It assumes the runtime architecture (deferred to Paper 5), the reading study (Paper 2), the evidence/trust model (Paper 3), pattern authoring (Paper 6), and diagnostics (Paper 8), and cross-references rather than re-explains them. Per the caveat canon (item 6), every empirical statement here is a hypothesis or a design, never a finding.

1.3 2. Background and related work

`prefers-reduced-motion` and platform motion signals. The web platform exposes a user preference for reduced motion as a CSS/JS media query, intended for readers for whom motion is a barrier rather than an enhancement. Fundamental reads it through an injected host capability — `reducedMotion()` on the `FieldHost` interface, documented as "whether the user prefers reduced motion (freezes the sim)" (`packages/core/src/engine/host.ts`) and implemented in the browser host via `matchMedia('(prefers-reduced-motion: reduce)')` (`packages/dom/src/browser-host.ts`). The standard's framing — motion is a *preference dimension*, not a binary on/off of the interface — is the seed of this paper's model. [prefers-reduced-motion]

WCAG motion guidance and vestibular accessibility. Accessibility guidelines treat motion as a hazard to be controllable and treat animation-from-interactions as something a user must be able to disable without losing function. The motivating clinical reality is vestibular dysfunction, for which motion — especially large, parallax, or unexpected motion — can induce nausea, dizziness, and disorientation. The design obligation that follows is not "remove motion" but "ensure no information or function is *lost* when motion is removed." [wcag-motion] [vestibular-accessibility; frederick2013]

Color is not the only carrier of meaning. A long-standing accessibility principle holds that color must never be the sole means of conveying information, because color perception varies across readers.

Fundamental generalizes the principle from color to *any single expressive channel* — color, glow, and critically motion — and encodes it as a rule: "Color is not the only carrier of meaning. Motion is optional. Reduced motion preserves state without travel" (visual-language §16; §5.2). We treat "motion is not the only carrier" as the exact analogue of "color is not the only carrier," and apply the same remedy: pair the expressive channel with a redundant, accessible one. [use-of-color]

Animation as information versus decoration. A separate line of work distinguishes *functional* animation (motion that communicates causality, continuity, or hierarchy) from *decorative* animation (motion that does not). Fundamental's truth-mode taxonomy (Paper 1, §6.3) gives this distinction teeth at the level of *forces*; this paper extends it to the level of *rendering*, asking of each motion: does it carry state that has no other carrier? If yes, the static equivalent must carry that state too; if no, it may simply be removed. [baecker1990; material-design-motion]

The degraded-fallback antipattern. A separately authored reduced-motion path risks divergence from the primary surface unless it is derived from the same state and tested against the same meaning contract. We position Fundamental against this directly: the contribution is not "Fundamental has a reduced-motion fallback" — many systems do — but that the reduced surface is *derived from the same state* and is *checkable against the primary surface*. This is an engineering-risk framing, not an externally established empirical claim.

The distinguishing stance, as in the rest of the family, is *epistemic*: Fundamental makes the accessibility property auditable — a guard that throws, lint that warns, a schema field that is required, a preview that renders the reduced surface — so the claim "meaning survives without motion" can be checked rather than asserted.

1.4 3. The equivalence model

The model traces a single chain from meaning to its renderings and asks what must be invariant along it. Read it as: *the meaning is fixed at the source; the renderings vary; the static rendering must lose nothing the meaning depended on.*

```
semantic source  →  visual-semantic binding  →  motion behavior   (motion allowed)
                                                 static equivalent (reduced motion)
```

1.4.1 3.1 Semantic source: live DOM is canonical

The meaning lives in the DOM, not in any rendering of it. A Fundamental page is *ordinary semantic HTML*: the Reading Field study is "an ordinary article — sections, headings, citations, a table of contents" wired to the platform (Paper 1, §5.4). The accessibility canon makes the source's primacy a rule: "Live semantic text remains in the DOM. Vectorized text must be paired with hidden or equivalent semantic text. Canvas visuals require semantic DOM fallback" (visual-language §16). Field state that is *derived* — attention, memory, relationship strength — is written back as CSS custom properties and typed registry state, not as new prose; the prose a reader (or a screen reader) consumes is unchanged by the field. The semantic source is what every other layer must preserve. It is canonical because it is the layer a reader can always reach: through the accessibility tree, through reader mode, through plain HTML with every visual layer stripped.

1.4.2 3.2 Visual–semantic binding: pair the expressive layer to its source

An expressive visual layer — an SVG overlay, a canvas render surface, a WebGL field — is bound to the semantic element it represents through the platform's `VisualBindingRegistry` (`packages/dom/src/visual-bindings.ts`). A binding declares a `visual`, an optional `semanticSource`, and a `role` (`decorative | representation | debug | relationship | measurement`), and carries an accessibility record: whether the visual is `aria-hidden`, whether its role *requires* a semantic source, and whether it would expose duplicate semantics to assistive technology. The registry encodes two rules in code, not prose:

- A visual whose role *represents* meaning (`representation`, `relationship`) **must** bind a semantic source; an orphan is an error.
- A visual that does *not* carry independent meaning **should** be `aria-hidden`, so screen readers do not read the decorative layer.

The canonical statement is the native primitive Fundamental wishes existed: "this visual represents that semantic element; don't double-expose it" (file header). The binding is the seam where the expressive layer is *attached* to meaning rather than *substituting* for it — which is what lets the expressive layer be freely transformed (including reduced to static) without endangering the meaning underneath.

1.4.3 3.3 Motion behavior and static equivalent

When motion is allowed, the bound visual renders field state as movement. When motion is not allowed, the *same field state* renders as a static equivalent. The mapping is not improvised per page; it is a fixed table in the Accessibility Contract (system-contracts §14), reproduced in the shipped accessibility preview (`apps/site/src/pages/docs/accessibility-preview.astro`):

Motion form (full mode)	Static equivalent (reduced mode)	State preserved
particles travel	particles freeze or fade	local density / occupancy
sparks	static highlight	a discrete event / threshold crossing
`fieldflow` ribbons	static field lines	relationship direction / routing
heat trails	soft wash	accumulated heat / activity
body motion	CSS state only	the body's own engaged/lit state
turbulence	contour snapshot	instantaneous field shape

Each row is a *representation swap*, not an information drop: the right column is a different rendering of the same scalar the left column animated. Travel becomes presence; a spark (a moment in time) becomes a highlight (the same moment, held); a flowing ribbon (direction shown over time) becomes a static field line (direction shown in space). The reduced render preset names this surface directly: `reduced` = "static contours + DOM state" (visualization-methods-taxonomy §0), as distinct from the animated presets (`beautiful`, `plasma`, `thermal`).

The emission channel shows the swap at the level of a single value. Under reduced motion, glow does not pulse — it flattens to a static highlight: `emission()` maps heat to radius with linear (not eased) interpolation and lowers the alpha ($0.8 \rightarrow 0.5$) when `reducedMotion` is set, so a pulsing bloom becomes a flat, bounded highlight (`packages/core/src/visual/channels.ts`). The state (heat) is identical; only its rendering loses its temporal dimension.

1.4.4 3.4 Three invariants

The chain is disciplined by three invariants, each restated from the accessibility canon and each enforced somewhere in code (§4):

1. **Meaning is never motion-only.** Every motion that carries state has a static equivalent that carries the same state. The contract states it twice — "Motion is optional. Meaning is never motion-only" (system-contracts §14) — and the pattern schema makes the equivalent *mandatory* (§4.2). This is the load-bearing invariant; the other two are its corollaries for specific channels.
2. **Color is never the sole carrier of meaning.** State that a reader must perceive is encoded redundantly — outline, icon, text, layout, or tone — not in hue alone (visual-language §5.2). The reduced surface, having dropped motion, must not then lean on color as a single replacement channel; that would trade one inaccessible monopoly for another.
3. **State still tracks under reduced motion; only easing drops.** The field keeps computing. The shipped Reading Field path is the canonical demonstration: under `prefers-reduced-motion` the CSS drops the transitions (`transition: none`) while the JavaScript still computes attention and writes `--field-attention`, so "the values simply snap instead of easing" (`apps/site/src/pages/docs/reading-field.astro`

the page's own copy: "The animation stops; the meaning does not. Attention and memory still track, the current section is still marked, the rails still fill"). Reduced motion removes the *interpolation between states*, not the states.

1.4.5 3.5 Grounding: the shipped accessible surfaces

The model is not purely normative; two shipped surfaces instantiate it. The **accessibility preview** (`apps/site/src/pages/docs/accessibility-preview.astro`) renders any composition as its reduced surface side-by-side with the motion-on surface, with the full→reduced mapping table; its headline reads its "lit" state from a single `--d` scalar that drives weight, color, and glow, with motion as a purely decorative pulse that both the OS setting and an in-page toggle disable — leaving the lit state fully readable. The **Reading Field** (`/docs/reading-field`, invariant 3 above) is the real-content instance. The flagship lists both as shipped (Paper 1, §9.1); they are the existence proofs that the chain of §3.1–§3.4 can be authored on real pages, not only specified.

1.5 4. The conformance model

The central contribution is not the equivalence chain — equivalence chains can be drawn for any fallback — but making the chain *checkable*. This section gives the mechanisms that turn "motion is not meaning" into a property a tool can fail. Three are shipped (lint, schema, guard); one is proposed (the automated equivalence harness); the preview is the human-in-the-loop verifier that ties them together.

1.5.1 4.1 Shipped: the accessibility lint rules

The platform's `lintPlatform()` (`packages/dom/src/lint.ts`) aggregates pure, read-only rules over the registries; three of them are accessibility seams. Lint *reads — it never mutates state, physics, or the DOM* (file header), so running it is side-effect free.

- **`visual-orphan`.** A visual binding whose role represents meaning but has no semantic source. This catches the precise failure where an expressive layer *becomes* the carrier of meaning instead of *representing* a carrier — exactly the condition under which reducing or hiding the visual would lose meaning. Surfaced from `VisualBindingRegistry.lint()` as `orphan-representation` (severity *error*) and mapped to `visual-orphan` in `lintVisuals()`.
- **`visual-not-hidden`.** A visual whose role is not `representation` and that is not `aria-hidden`. This catches the inverse failure: a layer with no independent meaning that nonetheless exposes itself to assistive technology, doubling what the screen reader reads. Severity *warning*.
- **`feedback-non-css-var`.** A feedback binding that writes ARIA or attributes instead of a `--field-*` custom property. Severity *error*. This rule is an accessibility rule in disguise: it enforces that field *state* is written to CSS variables (a presentational channel) and never masquerades as *accessibility state* — "state must not write ARIA/attributes" (`lintFeedbackVars`). Keeping field density out of the accessibility tree is what stops the field from spamming assistive technology with per-frame churn, which the contract forbids ("Field events should not spam assistive tech", system-contracts §14). The single-writer discipline — only the `FeedbackRegistry` turns state into DOM-visible output (Paper 1, §5.2) — is what makes this rule sufficient.

These rules are listed in the Platform-Lint Contract, marked **Implemented** (system-contracts §23), and surface alongside the structural rules (`relation-target-missing`, `state-unregistered`, `overlay-without-links`, `measurement-off-phase`). Their accessibility role is the §16 visual-language rule made operational (visual-language §18.1).

1.5.2 4.2 Shipped: the pattern schema makes a fallback required

A field pattern — the portable, serializable unit of field behavior (Paper 1, §7.3; developed in Paper 6) — cannot validate without declaring its reduced-motion equivalence. The schema (`packages/core/src/recipes/schema.ts`)

makes `accessibility` a non-optional field of `FieldPattern`, with the comment that states the policy: "the reduced-motion + meaning-without-motion equivalent — required: no pattern is motion-only." The field's shape is two required strings:

```
interface AccessibilityRecipe {
  /** what replaces motion under prefers-reduced-motion. */
  reducedMotion: string;
  /** how meaning survives without color/motion. */
  meaningWithoutMotion: string;
}
```

and the validator enforces both:

```
if (!r.accessibility || !r.accessibility.reducedMotion || !r.accessibility.meaningWithoutMotion)
  problems.push({ path: 'accessibility',
    issue: 'reducedMotion + meaningWithoutMotion are required (no pattern is motion-only)' });
```

This is the antipattern of §1.2 inverted at the authoring layer. A second, drifting fallback path is *possible* in a system where the fallback is optional; here it is structurally impossible to ship a valid pattern that has not said, in words, what replaces its motion and how its meaning survives without color or motion. The declaration is *prose* today (a human sentence), which is a real limitation we return to in §6 — the schema requires that the author *answer the question*, not yet that a machine *verify the answer*.

1.5.3 4.3 Shipped: the reduced-motion guard and the travel-gating user agent

Two further mechanisms enforce the invariants at runtime rather than at authoring time.

The **reduced-motion guard** (`packages/core/src/contracts/guards.ts`) is the assertion form of invariant 1. `assertReducedMotionFallback(hasFallback)` throws a `MISSING_REDUCED_MOTION` error — "motion-dependent meaning needs a reduced-motion fallback" — when a motion-dependent surface has no fallback declared. The named error is in the project's error taxonomy as "no accessible fallback" (system-contracts §17). The accessibility test set asserts the guard both ways: it *does not throw* with a fallback and *throws* without one (`packages/core/src/contracts/a11y.test.ts`).

The **travel-gating user agent** (`packages/core/src/agents/user-agent.ts`) is invariant 3 at the level of the input source. The user agent projects a field source from pointer, focus, and selection. Under reduced motion, the *moving* part of that source — the pointer "wake" — is dropped, while the *static, accessible* part — a focused element's attention well — remains:

```
wake: !u.reducedMotion && moving ? { x: u.px!, y: u.py!, vx: u.vx, vy: u.vy } : null,
focus: u.focusId,   // present even under reduced motion
```

This is the code form of the interaction model's rule "focus creates state, not travel" (interaction-and-relationship-model §10): the reduced surface keeps the accessible attention well that focus creates and drops the travel that pointer motion creates. The a11y test verifies it directly — "no travel under reduced motion" yet "accessible focus source remains" (`packages/core/src/contracts/a11y.test.ts`). The same test confirms the emission flattening of §3.3 (reduced-motion alpha strictly below motion-on alpha) and that color/glyph are not sole carriers (the core visual lint raises `color-only-meaning` and `glyph-only-text`, `packages/core/src/visual/lint.ts`).

Taken together: the Accessibility Contract is a *named, published* contract in the `CONTRACTS` catalog, "enforced by the reduced-motion guard, the UserAgent travel-gating, and the a11y lint rules, with a dedicated test set" (system-contracts §14). The accessibility property is, in the family's sense, *auditable* — it is checked by running tests, not by trusting prose.

1.5.4 4.4 Proposed: an automated equivalence-conformance check

What ships verifies *components* of equivalence — a fallback is declared, a representation has a source, glow flattens, focus survives. What does not yet ship is a single automated check that the *meaning-bearing state*

of a whole composition is present and readable with motion disabled. We propose it, framed as a design, not a result.

The proposed **equivalence-conformance check** would, given a composition and its visual bindings:

1. Render the composition twice — full motion and reduced motion (`reducedMotion: true`), reusing the path the pattern applier already exposes (`applyRecipe(..., { reducedMotion })`, `packages/dom/src/apply-recipe.ts`; exercised by `FieldLoopDemo.astro`).
2. Enumerate the meaning-bearing state for each measured body — the `StateRegistry` values written back as `--field-*` variables, plus the typed relationships — in *both* renders.
3. Assert *state equivalence*: every meaning-bearing value present under full motion is present, and reads to the same threshold class, under reduced motion. Travel and easing may differ; the state keys and their thresholded values may not.
4. Assert *carrier redundancy*: each state a reader must perceive is exposed through at least one non-motion, non-color-only channel (text, outline, icon, layout, tone) — the lint rules of §4.1 composed into a single per-composition assertion.
5. Assert *no orphaned meaning*: no `visual-orphan`, and every `representation` visual still resolves to live semantic text after the visual layer is hidden.

The check would slot into the conformance methodology the family already uses for forces (Paper 5) and the project's Conformance Contract, which already lists a "reduced-motion test" among required proofs (system-contracts §16). Its output is a pass/fail plus the list of states that failed to survive — a *diagnostic*, which is where it hands off to Paper 8. We claim only the design here; the harness is not built.

1.5.5 4.5 Implementation status

To be precise, in the family's house style:

Shipped (verifiable in the registries, the schema, the contracts catalog, and the tests):

- The accessibility lint rules `visual-orphan`, `visual-not-hidden`, `feedback-non-css-var` in `lintPlatform()` (`packages/dom/src/lint.ts`, `visual-bindings.ts`), and the core visual-lint rules `color-only-meaning`, `glyph-only-text`, `missing-reduced-motion` (`packages/core/src/visual/lint.ts`).
- The pattern schema's **required** `accessibility.reducedMotion` and `accessibility.meaningWithoutMotion` fields and their validation (`packages/core/src/recipes/schema.ts`).
- The reduced-motion guard `assertReducedMotionFallback` / `MISSING_REDUCED_MOTION` (`packages/core/src/contra`
  and the travel-gating user agent (`packages/core/src/agents/user-agent.ts`), both with the a11y test set (`packages/core/src/contracts/a11y.test.ts`).
- The `reducedMotion()` host capability that freezes the simulation (`packages/core/src/engine/host.ts`, `packages/dom/src/browser-host.ts`).
- The accessibility preview (`apps/site/src/pages/docs/accessibility-preview.astro`), the reduced-motion path of the Reading Field (`apps/site/src/pages/docs/reading-field.astro`), and the static narrative collapse (visualization-methods-taxonomy §13: under `prefers-reduced-motion` the narrative "collapses to a static preview that still names each layer and shows its current state").
- The `reduced` render preset ("static contours + DOM state", visualization-methods-taxonomy §0).

Proposed (a design in this paper, not shipped): the automated equivalence-conformance harness of §4.4 — a single check that the *whole* meaning-bearing state of a composition survives motion being disabled. The pattern schema currently requires the author to *state* the equivalence in prose; the harness would *verify* it.

1.6 5. Evaluation plan

The conformance model of §4 can verify that meaning-bearing state is *present* under reduced motion. It cannot verify that a *human* perceives the reduced surface as equivalent — that the meaning survives in the reader, not only in the registry. That is an empirical question, and this section is its protocol. Per the caveat canon (item 6), it reports **no results**; it is a plan.

The project's evaluable claim is stated plainly: *Does reduced-motion equivalence preserve meaning?* This is a lighter study than the reading and evidence studies (Papers 2–3), because its dependent variable is narrower: not "does the field help?" but "does the reduced surface lose anything the full surface conveyed?"

1.6.1 5.1 Design and conditions

A between- or mixed-subjects design with three conditions over the same content (a Reading Field article and an Evidence Field panel, so the claim is tested on both a reading and a trust surface):

- **C1 — full-motion field:** the composition as authored, motion allowed.
- **C2 — reduced-motion field:** the *same* composition under `prefers-reduced-motion` — the static equivalents of §3.3, the field still computing.
- **C3 — static baseline:** the semantic HTML with the field layer absent entirely (no visual–semantic binding, no reduced equivalent) — the floor against which both field surfaces are measured.

C2 is the surface under test; C1 is the full expressive surface it must not fall below; C3 isolates how much of any effect is the field at all versus the prose.

1.6.2 5.2 Primary test: non-inferiority

The core test is **non-inferiority of C2 relative to C1** on meaning-preservation measures. The hypothesis is *not* that reduced motion is *better* — it is that reduced motion is *not worse* on the meaning the interface is supposed to convey:

- **Comprehension:** accuracy on questions about content and structure (what does this section say; which claim does this source support).
- **Orientation:** "where am I / where have I been" — can the reader locate the current section and recall which regions they have visited (the memory field, rendered statically under C2).
- **Recall:** delayed recall of content and of relationships (which claim was contradicted, which section was emphasized).

Non-inferiority requires a pre-registered margin: C2 is accepted as preserving meaning if its mean on each measure is within margin δ of C1 (one-sided test against the inferiority bound, not a two-sided test for a null difference). The margin and sample size are set a priori from a pilot; we do not state numbers here because no pilot has run.

1.6.3 5.3 Secondary measures

- **Preference among motion-sensitive readers.** Among participants who self-report motion sensitivity or who run with reduced motion enabled, *preference* and *self-reported comfort/symptoms* for C1 versus C2. The hypothesis is a comfort advantage for C2 with no comprehension cost — the whole point of the equivalence.
- **Motion-only meaning probe.** A targeted probe for any state a reader can extract *only* from motion: items answerable in C1 but not C2 would falsify the equivalence claim for that state and identify a static equivalent that is doing too little. This is the most important secondary measure, because it is the one that can *break* the central claim; a clean equivalence predicts no C1-exclusive items.
- **Task time and effort,** as a calibration that any C2 comfort gain is not bought with a hidden comprehension or speed penalty.

1.6.4 5.4 What a result would and would not show

A non-inferiority result on §5.2 plus a null motion-only-probe (§5.3) would support the claim that *this composition's* reduced surface preserves meaning. It would *not* establish equivalence for arbitrary compositions — that depends on the authored static equivalent (§6) — and it would not, on its own, separate the field's contribution from the prose's (which is why C3 is included). As throughout the family, the study is designed to be *falsifiable*: a C1-exclusive item, or a C2 comprehension deficit beyond δ, is a result that would send the static equivalents back to authoring.

1.7 6. Limitations

We state the model's limits plainly, as the caveat canon requires.

Equivalence is only as good as the authored static equivalent. The model guarantees that an author must *declare* a static equivalent and that the platform will *render* one; it does not guarantee that the declared equivalent actually captures the state the motion conveyed. A lazy equivalent — "particles freeze" with no thought to *which* state the travel encoded — passes the schema's required-field check and the orphan lint, yet may strand meaning. The schema requires an answer; it does not grade it. This is the gap the §5.3 motion-only probe and the §4.4 harness are designed to expose, and it is why the conformance model and the study are *complementary*, not redundant.

The conformance harness checks presence, not perceived equivalence. Even fully built, the proposed §4.4 check verifies that meaning-bearing *state* is present and readable with motion disabled — a necessary condition. It cannot verify that a human reads the static surface as *meaning the same thing*. Presence of a `--field-attention` value is not the same as a reader perceiving emphasis. Perceived equivalence is exactly what §5 must measure; no static analysis substitutes for it.

Prose declarations are weakly checkable. Today's `meaningWithoutMotion` is a human sentence. It makes the author answer the right question and documents the intent, but a machine cannot yet check that the sentence is *true of the rendered surface*. Tightening it toward a structured, checkable declaration (state keys preserved, redundant carriers named) is future work and a precondition for fully automating §4.4.

Generalization. The model is developed and grounded on the surfaces Fundamental ships well — long documents, evidence panels, dashboards (Paper 1, §8.6) — where state is relational and has natural static renderings (contours, highlights, field lines). On surfaces whose meaning is *intrinsically* temporal (a genuine animation of a process unfolding over time), "freeze the motion, keep the state" may have no faithful static equivalent, and the honest answer is a static *summary* plus the temporal content offered through another accessible channel, not a claim of equivalence. The model bounds where equivalence is achievable; it does not assert it everywhere.

1.8 7. Discussion

Why this matters for Fundamental. The two failure modes of §1.1 are the two ways an expressive field system dies: dismissed as decorative, or rejected as inaccessible. Treating motion as one representation of state defuses both at once. Because the static equivalent renders the *same* computed field, the motion is demonstrably *not* decoration — removing it provably loses nothing the state did not, and the conformance model is what makes "provably" literal. And because the reduced surface is the same field differently revealed rather than a separate, lesser path, it does not drift: there is one source of meaning, rendered two ways, with tooling that checks the renderings agree.

Why the principle generalizes. The move here — *motion is a representation of state, not the source of meaning* — is not specific to fields. Any interface that animates can ask of each motion: what state does this travel encode, and what is its static rendering? Systems that can answer get a reduced-motion mode for free (it is the static rendering of the same state) and a conformance story for free (the two renderings must agree on the state). Systems that *cannot* answer have learned something important: their motion was decoration, or worse, their motion was the *only* carrier of a meaning that should have had a redundant channel all along. The accessibility canon's "color is not the only carrier" becomes, generalized, *no single channel is the only carrier* — and motion is just the channel the literature warns about most.

Relationship to the rest of the family. The conformance posture here is the same as the flagship's truth modes and passports (Paper 1, §6) and the pattern conformance gate (Paper 6): make the honest boundary mechanical rather than rhetorical. Accessibility is, in this view, not a separate concern bolted

onto the runtime but another property the runtime is built to make *checkable* — the same discipline, pointed at reduced motion.

1.9 8. Conclusion

Motion-rich interfaces face a real double bind: expressive enough to matter is expressive enough to exclude, yet safe enough to include can look like decoration that never mattered. Fundamental's stance — *motion is one representation of state, not the source of meaning* — dissolves the bind by making reduced motion the same field differently revealed: the field keeps computing, only the easing drops, and every travelling effect has a static equivalent that renders the same state. The contribution of this paper is to make that equivalence a *testable conformance property* rather than a hand-waved fallback. The shipped mechanisms — the `visual-orphan` / `visual-not-hidden` / `feedback-non-css-var` lint rules, the pattern schema's required reduced-motion-and-meaning-without-motion fields, the reduced-motion guard, the travel-gating user agent, and the accessibility preview — already enforce the components of equivalence; the proposed equivalence-conformance harness would check the whole. We have been explicit that this verifies *presence* of meaning-bearing state, not human-*perceived* equivalence, which is why the evaluation plan asks, as a falsifiable non-inferiority study, whether the reduced surface preserves meaning for real readers. Motion can be expressive without being load-bearing — and a field system can prove it.

1.10 Appendix A. Reproducibility

Every accessibility claim in this paper is checkable against the repository. The load-bearing anchors:

- **Lint rules:** `packages/dom/src/lint.ts` (`lintVisuals`, `lintFeedbackVars`, `lintPlatform`), `packages/dom/src/visual-bindings.ts` (`VisualBindingRegistry.lint()`, `orphan-representation` / `visual-not-hidden`), `packages/dom/src/overlays.ts`, `packages/core/src/visual/lint.ts` (`color-only-meaning`, `glyph-only-text`, `missing-reduced-motion`).
- **Recipe-schema accessibility field:** `packages/core/src/recipes/schema.ts` (`AccessibilityRecipe`, the required `accessibility` field on `FieldPattern`, and the validator clause forbidding a motion-only pattern).
- **Reduced-motion guard, travel-gating, host capability:** `packages/core/src/contracts/guards.ts` (`assertReducedMotionFallback`, `MISSING_REDUCED_MOTION`), `packages/core/src/agents/user-agent.ts` (`userFieldSource` wake-gating), `packages/core/src/visual/channels.ts` (`emission` reduced-motion flattening), `packages/core/src/engine/host.ts` / `packages/dom/src/browser-host.ts` (`reducedMotion()`), `packages/core/src/contracts/index.ts` (Accessibility Contract in `CONTRACTS`).
- **Tests:** `packages/core/src/contracts/a11y.test.ts` (fallback required; meaning survives without motion; color/glyph not sole carriers; events thresholded; contract published).
- **Accessibility preview and reduced-motion demo paths:** `apps/site/src/pages/docs/accessibility-preview.` (side-by-side + mapping table), `apps/site/src/pages/docs/reading-field.astro` (state-tracks / easing-drops path), `apps/site/src/components/FieldLoopDemo.astro` (`applyRecipe(..., { reducedMotion })`), `apps/site/src/pages/docs/accessibility.astro`.

The canonical design documents corroborate the framing: the Accessibility Contract, Feedback Contract, Visualization Contract, Visual Language Contract, and Platform-Lint Contract (`docs/canonical/system-contracts.md` §§10, 11, 14, 19, 23); the Accessibility and Semantic Layer, the color accessibility rules, the Visual Contracts, and the Visual Binding/Overlay registries + Accessibility Preview (`docs/canonical/visual-language-and-geometry.md` §§5.2, 16, 18, 18.1, 18.2); the `reduced` preset and narrative-reveal collapse (`docs/canonical/visualization-methods-taxon` §0, §13); and "focus creates state, not travel" (`docs/canonical/interaction-and-relationship-model.md` §10).

1.11 Appendix B. Conversion notes (markdown → preprint)

Notation is kept LaTeX-compatible (the non-inferiority margin δ and the inline math translate directly). Figures referenced in prose but not yet drawn — the equivalence chain (§3, semantic source → binding → motion / static equivalent), the motion→static mapping table rendered as a figure (§3.3), and the three-condition study design (§5.1) — are produced at conversion time. External citations marked citation keys in `references.md` are resolved and verified before submission — never fabricated.

1.12 Citation coverage

External references this paper relies on, resolved into `references.md` where a real source supports the claim. Unsourced engineering-risk prose is labeled as such rather than cited:

- `[prefers-reduced-motion]` — the `prefers-reduced-motion` media-query specification (CSS Media Queries Level 5 / platform docs). *(resolved: [prefers-reduced-motion].)*
- WCAG 2.x motion / animation-from-interaction guidance (the success criteria governing motion-from-interaction and disabling non-essential animation). Resolved: [wcag-motion].
- Vestibular-disorder accessibility literature (clinical and HCI accounts of motion as a barrier: nausea, dizziness, disorientation from large/parallax/unexpected motion). Resolved: [vestibular-accessibility; frederick2013].
- "Color is not the only means of conveying information" / redundant-coding guidance (the WCAG use-of-color principle this paper generalizes from color to motion). Resolved: [use-of-color].
- Functional-versus-decorative animation in interface design (motion that communicates causality, continuity, or hierarchy versus motion that does not). Resolved: [baecker1990; material-design-motion].
- The degraded-fallback / dual-codebase-drift antipattern in accessibility practice (separate, under-tested accessible paths diverging from the primary surface). No canonical source is asserted; the paper frames this as an engineering-risk limitation.
- Non-inferiority trial design and margin selection (the statistical framing of §5.2, for the evaluation methodology). Resolved: [piaggio2012].